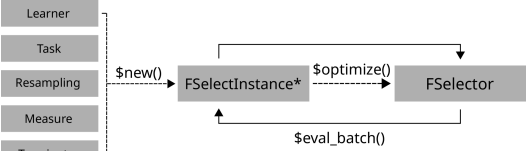


# Feature Selection with mlr3fselect::CHEAT SHEET

## Class Overview

The package provides a set of R6 classes which allow to (a) define general feature selection instances; (b) run black-box optimizers; (c) combine learners with feature selection (for nested resampling).



[NB: In many table prints we suppress cols for readability.]

## Terminators - When to stop

Construction: `trm(.key, ...)`

- `evals(n_evals)`  
After iterations.
- `run_time(secs)`  
After training time.
- `clock_time(stop_time)`  
At given timepoint.
- `perf_reached(level)`  
After performance was reached.
- `stagnation(iters, threshold)`  
After performance stagnated.
- `stagnation_batch(n, threshold)`  
After performance stagnated for batches.
- `combo(list_of_terms, any=TRUE)`  
Combine terminators with AND or OR.

```
as.data.table(mlr_terminators) # list all
```

Lists all available terminators.

## FSelectInstance\* - Search Scenario

Evaluator and container for resampled performances of feature subsets. The (internal) function `eval_batch(xdt)` calls `benchmark()` to evaluate a table of feature subsets. Stores archive of all evaluated feature subsets and the final result.

```
instance = fsi(task, learner, resampling, measure, terminator)
```

`store_benchmark_result = TRUE` to store resampled evals and `store_models = TRUE` for fitted models.

## Example

```
instance = fsi(task, learner, resampling, measure, terminator)
fselector = fs("random_search", batch_size = 10)
fselector$optimize(instance)
instance$result
# >   Petal.Length Petal.Width Sepal.Length Sepal.Width classif.ce
# > 1:      FALSE      TRUE      TRUE      TRUE      0.06
```

Use `FSelectInstanceBatchMultiCrit` for multi-criteria feature selection.

## FSelector - Search Strategy

Generates feature subsets and passes to instance for evaluation until termination. Creation: `fs(.key, ...)`

- `random_search(batch_size)`  
Random search.
- `exhaustive_search(max_features)`  
Exhaustive Search.
- `sequential(strategy)`  
Sequential Selection.
- `rfe(feature_fraction, recursive)`  
Recursive Feature Elimination.
- `design_points(batch_size, design)`  
User supplied feature subsets.

```
as.data.table(mlr_fselectors) # list all
```

Lists all available feature selection algorithms.

## Logging and Parallelization

```
lgr::get_logger("mlr3/bbotk")$set_threshold("<level>")
```

Change log-level only for mlr3fselect.

```
future::plan(strategy)
```

Sets the parallelization backend. Speeds up feature selection by running iterations in parallel.

## Executing the Feature Selection

```
fselector$optimize(instance)
as.data.table(instance$archive)
## >   Petal.Length Petal.Width Sepal.Length Sepal.Width classif.ce
## > 1:      TRUE      TRUE      TRUE      TRUE  0.09333333
## > 2:      TRUE      TRUE      TRUE      FALSE 0.09333333
instance$result # datatable row with optimal feature subset and estimated perf
```

Get evaluated feature subsets and performances; and result.

```
task_selected$select(instance$result_feature_set)
```

## Example

```
instance = fselect(fs("random_search"), task = tsk("iris"), learner = learner,
  resampling = rsmpl("holdout"), measures = msr("classif.ce"), term_evals = 20)
```

Use `fselect()`-shortcut.

## AutoFSelector - Select before Train

Wraps learner and performs integrated feature selection.

```
afs = auto_fselector(fselector = fselector,
  learner = learner, resampling = resampling,
  measure = measure, term_evals = 20)
```

Inherits from class `Learner`. Training starts feature selection on the training set. After completion the learner is trained with the “optimal” feature subset on the given task.

```
afs$train(task)
afs$predict(task, row_ids)
```

```
afs$learner
```

Returns learner trained on full data set with optimized feature subset.

```
afs$fselect_result
# >   Petal.Width Sepal.Length Sepal.Width classif.ce
# > 1:      TRUE      TRUE      TRUE      0.02
```

Access feature selection result.

## Nested Resampling

Just resample `AutoFSelector`; now has inner and outer loop.

## Example

```
inner = rump("holdout")
afs = auto_fselector(fs("random_search"), learner, inner, measure, term_evals = 20)
outer = rump("cv", folds = 2)
rr = resample(task, afs, outer, store_models = TRUE)
```

```
as.data.table(rr)
## >   learner      resampling iteration      prediction
## > 1: <AutoFSelector[38]> <ResamplingCV[19]> 1 <PredictionClassif[19]>
## > 2: <AutoFSelector[38]> <ResamplingCV[19]> 2 <PredictionClassif[19]>
```

```
extract_inner_fselect_results(rr)
# >   iteration Petal.Width Sepal.Length Sepal.Width classif.ce
# > 1:      1      TRUE      TRUE      TRUE      0.04
# > 2:      2      TRUE      TRUE      FALSE      0.00
```

Check inner results for stable features.

```
rr$score()
# >   learner iteration      prediction classif.ce
# > 1: <AutoFSelector[40]> 1 <PredictionClassif[19]> 0.02666667
# > 2: <AutoFSelector[40]> 2 <PredictionClassif[19]> 0.08000000
```

Predictive performances estimated on the outer resampling.

```
extract_inner_fselect_archives(rr)
# >   iteration Petal.Width Sepal.Length Sepal.Width classif.ce
# > 1:      1      FALSE      TRUE      FALSE      0.36
# > 21:      2      FALSE      TRUE      FALSE      0.44
```

All evaluated feature subsets.

```
rr$aggregate()
# > classif.ce
# > 0.05333333
```

Aggregates performances of outer resampling iterations.

```
rr = fselect_nested(fs("random_search"), task,
  learner, inner, outer, measure, term_evals = 20)
```

Use shortcut to execute nested resampling.